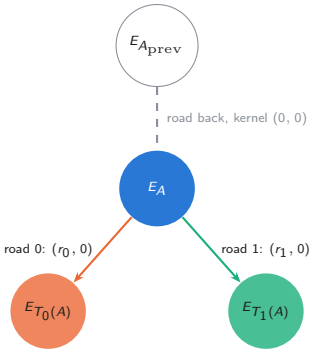
Isogenies on Oasis

Qiskit Fall Fest 2026, Track 1

Three Qiskiteers

November 2026

The campus



- A spot is a curve $E_A: \; y^2 = x^3 + Ax^2 + x$, with $A \in \mathbb{F} = \mathbb{F}_p(i)$, $p \equiv 3 \pmod 4$.
- A road is a 2-isogeny. Its kernel is one 2-torsion point $(r, 0)$:

$$r_{0,1} = \frac{-A \pm \sqrt{A^2 - 4}}{2}, \quad \text{sorted by } \operatorname{key}(u + vi) = u + pv.$$

- Bit b takes the road through r_b:

$$T_b(A) = \operatorname{can}(2 - 4r_b^2), \quad \operatorname{can}(z) = \text{the smaller key of } \pm z.$$

- A password is a route $x = b_1 \cdots b_L$. It ends at

$$H(x) = j(E_{A_L}) = \frac{256(A_L^2 - 3)^3}{A_L^2 - 4}.$$

- Small campus: $p = 1019$, $A_0 = 977 + 297i$, $L = 12$, $j^* = 797 + 921i$.

Terms

The campus

- **Spot**: a curve E_A, a vertex of the graph.
- **Road**: a 2-isogeny, an edge. Each spot has 3 roads. A route never goes back along the road it arrived by, so each step has 2 choices.
- **Route**, **password**: the bits $x = b_1 \cdots b_L$; bit b_t picks the road at step t.
- **Maths department**: the start spot A_0.
- **ANC**: the target of Part I, every spot with $j(E_A) = j^*$.
- **Clock Tower**: the target of Part II, every spot with $j(E_A) \in \mathbb{F}_p$.

The method

- **Survey**: a classical map of the campus, built from p alone by walking each road once. Part I builds its circuit from it.
- **Query**: one use of the one-step circuit.
- $\{cx, rz, sx, x\}$: the IBM basis gate set we transpile to.
- **Djinn**, **volunteer** (Part IV): the spot names are hidden behind random labels. The Djinn returns the label at the end of a road; the volunteer reports whether a labelled spot is on the Clock Tower or at ANC.

The four parts

Part I **Every password.** On the small campus ($p = 1019$, $L = 12$), list every route from A_0 to ANC with a quantum procedure, give the confidence that the list is complete, and count the queries.

Part II **Reach the Clock Tower.** For a general prime p, find a route of length $L = O(\log p)$ from A_0 that ends on the Clock Tower. Give the quantum cost as a function of p (an upper and a lower bound), compare it with classical search, and check it on data.

Part III **The step from arithmetic.** Build the one-step circuit $|A, b\rangle\,|0\rangle \mapsto |A, b\rangle\,|T_b(A)\rangle$ from field arithmetic alone, without the survey. Check it on every input at $p = 1019$, and count its cost at $p \approx 2^{256}$.

Part IV **Hidden names.** The spot names are replaced by random labels (the Djinn model). Say which methods still work, which stop working, and what lower bound holds.

Constraints

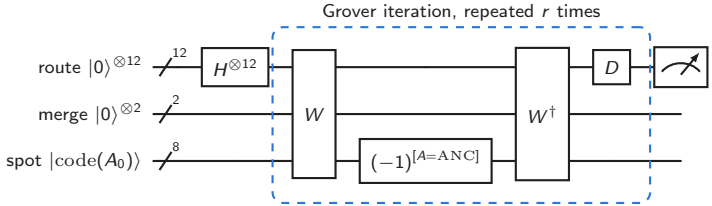
Search rules

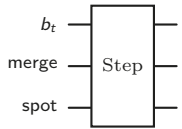
- The survey walks each road once (at most 2 moves per spot). It does not use A_0, L or j^*.
- After the survey: no path counting, no search backwards from ANC, no meet-in-the-middle, no route lists.
- The only classical work: walk one measured route once, to check it.
- The ANC test marks the codes of the ANC spots with a lookup on the spot register, inside the circuit. It does not evaluate H.
- If no route reaches ANC, the program reports it.

Build rules

- All circuits are built and run in Qiskit.
- Resources are given after transpiling to a stated basis: queries, qubits, two-qubit gates, depth. Part III also gives the Toffoli and T counts.
- One command reproduces every result. It takes p, A_0, L, j^*.
- One query is one use of the one-step circuit.

Part I: the circuit



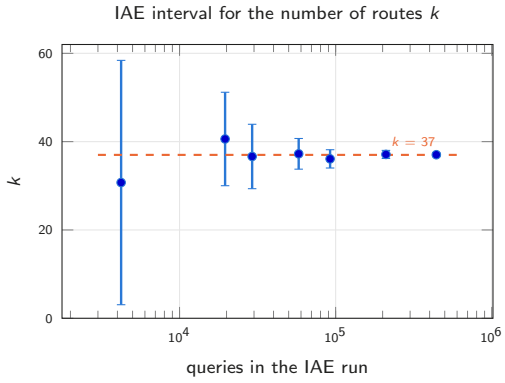


$W = \mathrm{Step}_{12} \cdots \mathrm{Step}_1$, step t reads bit b_t

- Survey: 255 spots, 510 roads. Each spot gets an 8-bit code.
- Two roads end at the same spot ($A = 6$), so the step alone is not one-to-one. Two merge qubits record which road arrived; the step is then a permutation of basis states.
- A single-target gate flips one bit, controlled by the other bits. Every permutation of n bits is a product of at most $2n - 1$ of them (De Vos and Van Rentergem, 2008). Our step uses 17.
- Oracle: W, phase -1 on the ANC codes, $W^\dagger$; 24 queries. $D = 2\,|s\rangle\langle s| - I$ is the Grover diffusion, $|s\rangle$ the uniform superposition.
- 22 qubits. In $\{cx, rz, sx, x\}$: one step 13,395 CX, depth 42,173.

L. K. Grover, *A fast quantum mechanical algorithm for database search*, STOC 1996.
A. De Vos, Y. Van Rentergem, *Young subgroups for reversible computers*, Adv. Math. Commun. 2 (2008).

Part I: counting the routes



- **IAE** (Iterative Amplitude Estimation): it estimates $a = k/N$ from circuits $Q^m A|0\rangle$ with Grover powers Q^m. Each round makes the confidence interval smaller (Appendix A1).
- We stop when the interval for $k = aN$ holds one integer. Result: $k = 37$ at confidence ≥ 0.99.
- $k = 0$ gives an interval at 0, and the program reports that no route reaches ANC.
- **Why IAE.** We first tried maximum-likelihood amplitude estimation. With few shots its likelihood had a second peak at $k \pm 1$ ($k = 5$ against 6 at $N = 256$). IAE keeps $(2m + 1)\theta_a$ inside one half-period, so this cannot happen. Counting by phase estimation needs a quantum Fourier transform and an extra register; IAE needs neither.

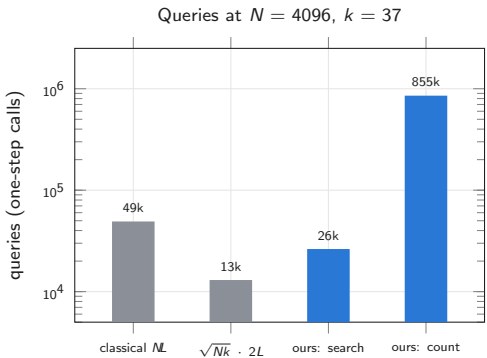
D. Grinko, J. Gacon, C. Zoufal, S. Woerner, *Iterative quantum amplitude estimation*, npj Quantum Inf. 7 (2021).
Y. Suzuki et al., *Amplitude estimation without phase estimation*, Quantum Inf. Process. 19 (2020).
G. Brassard, P. Høyer, A. Tapp, *Quantum counting*, ICALP 1998.

Part I: finding every route

- Grover with $r = 8$ iterations per run, 137 runs. Each measured route is checked once by one classical walk.
- A run succeeds with probability just under 1, so it can return a route that misses ANC. The check caught 1 such run, and it was discarded.
- All 37 routes are found. They end at 3 spots:

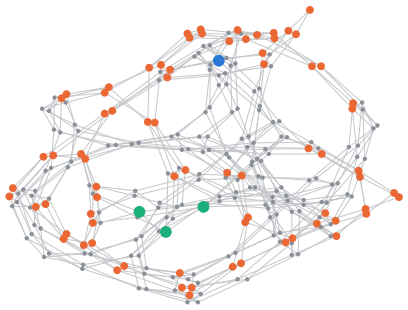
A_{12}	routes	j
$3 + 10i$	14	$797 + 921i$
$933 + 205i$	9	$797 + 921i$
$561 + 411i$	14	$797 + 921i$

- Chart: one-step calls. Grey: walk all N routes classically (NL), and the best possible quantum search for all k routes ($\sqrt{Nk} \cdot 2L$). Blue: our run.
- Our search is within $2\times$ of the best possible. The count costs the most: at $N = 4096$ the constants of IAE dominate (32 shots per round, the confidence terms). For large N the scaling, $\sqrt{Nk}$ against N, decides.



J. van Apeldoorn, S. Gribling, H. Nieuwboer, *Basic quantum subroutines: finding multiple marked elements and summing numbers*, Quantum 8 (2024).

Part II: the target



$p = 1019$: 255 spots, one connected graph. ● on the tower (75), ● ANC (3), ● A_0

- Clock Tower: $T_p = \{E : j(E) \in \mathbb{F}_p\}$.
- Deuring–Eichler theorem (counts the supersingular curves defined over $\mathbb{F}_p$): $|T_p| = 2h(-p)$ or $h(-p)$. Here $h(-p)$ is the class number of $\mathbb{Q}(\sqrt{-p})$.
- Siegel's theorem (growth of class numbers): $h(-p) = p^{1/2+o(1)}$. All spots: $|V_p| = p/12 + O(1)$.

$$\frac{|T_p|}{|V_p|} = p^{-1/2+o(1)}$$

- Small campus: 75 of 255 spots (29%). At $p \approx 2^{256}$: about 1 spot in 2^{125}.

M. Deuring, *Die Typen der Multiplikatorenringe elliptischer Funktionenkörper*, Abh. Math. Sem. Hamburg 14 (1941).
C. Delfs, S. D. Galbraith, *Computing isogenies between supersingular elliptic curves over $\mathbb{F}_p$*, Des. Codes Cryptogr. 78 (2016).
C. L. Siegel, *Über die Classenzahl quadratischer Zahlkörper*, Acta Arith. 1 (1935).

Part II: from spots to routes

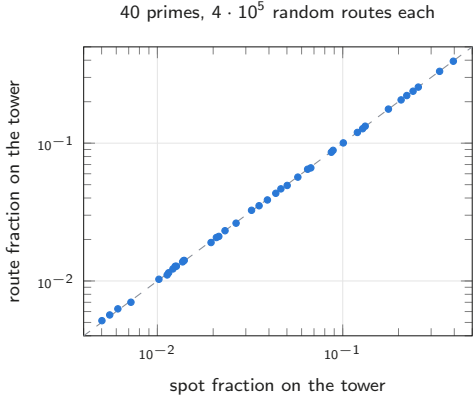
- The graph is connected (from any spot, the survey reaches all 255), 3-regular and Ramanujan: every nontrivial eigenvalue has $|\lambda| \le 2\sqrt{2}$, the best possible expansion (Pizer, 1990). A route is a walk on it without backtracking.
- Lubetzky–Peres (how fast such walks mix on Ramanujan graphs): the end of the walk is close to uniform in total variation distance,

$$d_{\mathrm{TV}} \le O\big(\sqrt{p}\, 2^{-L/2} L\big).$$

- With $L = \lceil 3 \log_2 p \rceil$ this is $\widetilde{O}(p^{-1})$, much less than the target fraction. So

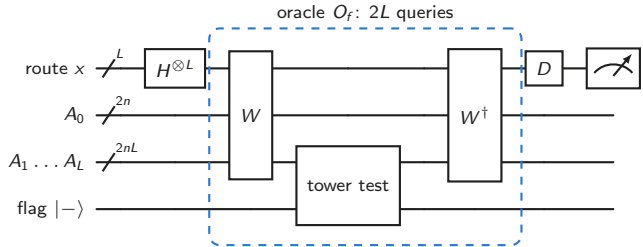
$$q_L = \Pr_x[f(x) = 1] = p^{-1/2+o(1)}.$$

- Data (right): route fraction = spot fraction, for 40 primes, already with $L = 2\lceil \log_2 p \rceil$.



A. K. Pizer, *Ramanujan graphs and Hecke operators*, Bull. AMS 23 (1990).
E. Lubetzky, Y. Peres, *Cutoff on all Ramanujan graphs*, GAFA 26 (2016).

Part II: the upper bound



- $O_f |x\rangle = (-1)^{f(x)} |x\rangle$, $f(x) = [j(E_{A_L(x)}) \in \mathbb{F}_p]$.
- With $\sin^2\theta = q_L$: after t iterations, $P(t) = \sin^2((2t+1)\theta)$.
- $t \approx \frac{\pi}{4\sqrt{q_L}} = p^{1/4+o(1)}$ iterations, each $2L = O(\log p)$ queries.
- q_L unknown: BBHT (Boyer–Brassard–Høyer–Tapp) runs Grover with random iteration counts; at most $\frac{9}{4}\sqrt{1/q_L}$ iterations on average.
- Classical random routes: $1/q_L = p^{1/2+o(1)}$.

M. Boyer, G. Brassard, P. Høyer, A. Tapp, *Tight bounds on quantum searching*, Fortschr. Phys. 46 (1998).

Part II: the lower bound

Black-box model

Input: an unknown $f : \{0,1\}^L \to \{0,1\}$ with k marked inputs. The only access is the oracle $O_f |x, z\rangle = |x, z \oplus f(x)\rangle$.

BBBV (Bennett–Bernstein–Brassard–Vazirani) and Zalka: every quantum algorithm with constant success needs

$$\Omega\left(\sqrt{N/k}\right) = \Omega\left(q_L^{-1/2}\right) = \Omega\left(p^{1/4-o(1)}\right)$$

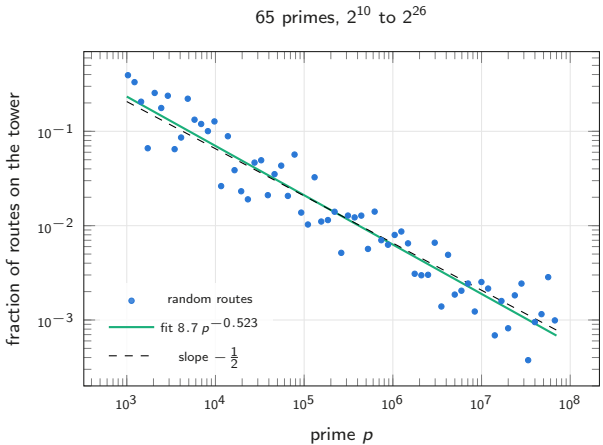
oracle queries. Grover meets this bound.

The real arithmetic

- The step is explicit field arithmetic. An algorithm can use this structure, so the black-box bound does not apply.
- Deciding whether a route to the tower exists is in NP (Appendix A3). BQP is the class of problems a quantum computer solves in polynomial time. A proof that no polynomial-time quantum algorithm exists would show NP $\not\subseteq$ BQP, and so P $\neq$ NP.
- No known algorithm beats $p^{1/4}$ under these rules (Appendix A4).

C. H. Bennett, E. Bernstein, G. Brassard, U. Vazirani, *Strengths and weaknesses of quantum computing*, SIAM J. Comput. 26 (1997).
C. Zalka, *Grover's quantum searching algorithm is optimal*, Phys. Rev. A 60 (1999).

Part II: data



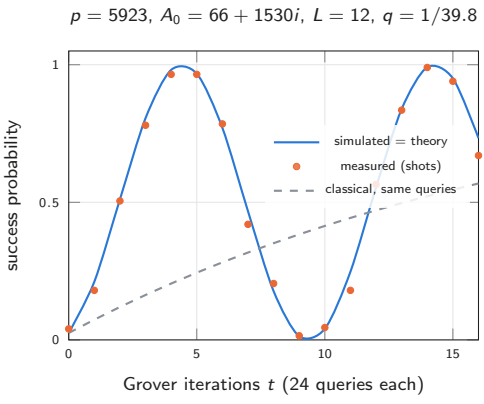
- Fit: $\widehat{q} \approx 8.7\,p^{-\alpha}$,

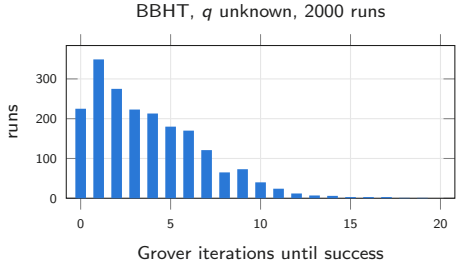
$$\widehat{\alpha} = 0.523 \pm 0.040$$

(65 primes, 2 standard errors).

- Exact spot fractions, 40 primes: $\widehat{\alpha} = 0.560 \pm 0.081$.
- The scatter comes from the class number $h(-p)$, not from noise: the same primes are high in both columns.
- Theory: $\alpha = \frac{1}{2}$.

Part II: demonstration

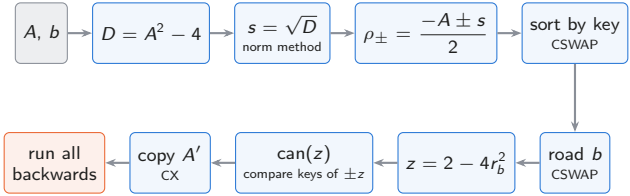




- Simulation = theory to 4 decimals. At $t = 4$: 0.981, against 0.205 classically.
- BBHT: on average 91 queries and 7.9 classical route checks. Classical random routes: 477 steps.
- 25 qubits. One iteration in $\{cx, rz, sx, x\}$: $3.40 \cdot 10^6$ CX, depth $1.11 \cdot 10^7$.

M. Boyer, G. Brassard, P. Høyer, A. Tapp, *Tight bounds on quantum searching*, Fortschr. Phys. 46 (1998).

Part III: one step from arithmetic



Square root from the norm

$D = \alpha + \beta i$. Two exponentiations in $\mathbb{F}_p$:

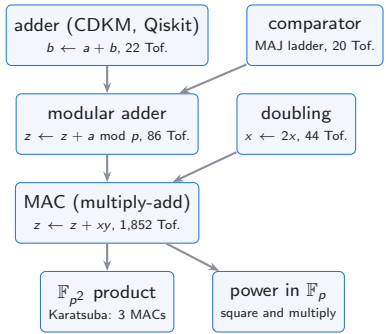
$$\lambda = (\alpha^2 + \beta^2)^{\frac{p+1}{4}}, \quad \delta = \tfrac{\alpha+\lambda}{2}, \quad r = \delta^{\frac{p-3}{4}}.$$

$y = r\delta$, $t = \beta r/2$, $q = [yr = 1]$: $s = y + ti$ if $q = 1$, else $s = -t + yi$. No inversion (Appendix A5).

- No branches: each condition is computed into a flag qubit, which controls the gates. CSWAP is a controlled swap.
- Any root s works: the key sort fixes the order of $\rho_\pm$.
- $|A, b\rangle|0\rangle \mapsto |A, b\rangle|T_b(A)\rangle$. Bennett's method returns all work qubits to 0: compute, copy A', run the computation backwards.
- Clock-Tower test without division: write $j = N/D$ and $\overline{a + bi} = a - bi$. Then $j \in \mathbb{F}_p \iff \overline{N}D = N\overline{D}$.

C. H. Bennett, *Logical reversibility of computation*, IBM J. Res. Dev. 17 (1973).

Part III: building blocks



Toffoli counts for $p = 1019$, $n = 10$ bits

- Gates: X, CX, Toffoli, CSWAP only.
- CDKM: the Cuccaro–Draper–Kutin–Moulton ripple-carry adder, built into Qiskit. The comparator keeps only its carry chain (MAJ gates).
- Modular adder: add, subtract p, add p back if the sign bit is 1, then clear the flag with the comparator.
- Doubling is a relabelling of qubits plus one subtraction of p.
- MAC: for each bit y_i, add $y_i \cdot x$, then double x. This gives n modular adders, so $O(n^2)$.
- Karatsuba: an $\mathbb{F}_{p^2}$ product from 3 products in $\mathbb{F}_p$ instead of 4.
- An exponentiation is $O(n)$ products, so one step is $O(n^3)$.

S. A. Cuccaro, T. G. Draper, S. A. Kutin, D. P. Moulton, *A new quantum ripple-carry addition circuit*, arXiv:quant-ph/0410184 (2004).
A. Karatsuba, Yu. Ofman, *Multiplication of multidigit numbers on automata*, Soviet Physics Doklady 7 (1963).

Part III: checks at $p = 1019$

Every input

- Qiskit Aer's matrix-product-state simulator, 587 qubits.
- Step: 510 inputs (255 spots, both bits). Output = classical $T_b(A)$, inputs unchanged, every work qubit 0. **510/510.**
- Clock-Tower test: **255/255** spots.
- 32 inputs on random spots at $p = 46351$: all correct.

Why basis states are enough

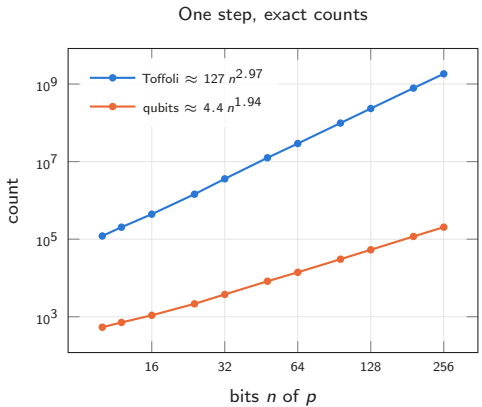
X, CX, Toffoli and CSWAP send basis states to basis states, so the circuit U is a permutation matrix. If $U|x\rangle|0\rangle = |x\rangle|f(x)\rangle$ for every x, then by linearity

$$U \sum_x c_x |x\rangle|0\rangle = \sum_x c_x |x\rangle|f(x)\rangle .$$

Phases

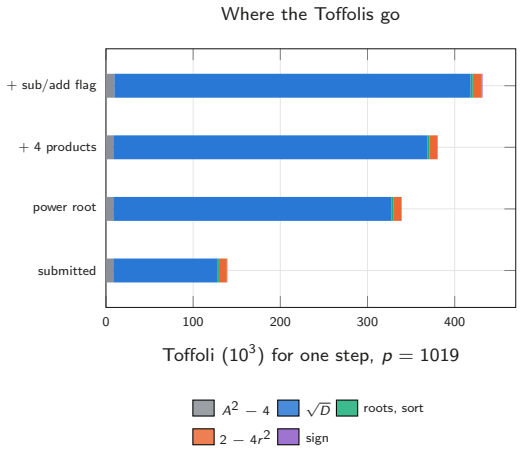
64 states $\cos w\,|x\rangle + e^{i\phi} \sin w\,|y\rangle$ go through the circuit and then the inverse of a reference circuit (Appendix A7). All return to $|0\rangle$. A deliberate phase error is detected.

Part III: cost at $p \approx 2^{256}$



one step, $n = 256$	
Toffoli	$1.83 \cdot 10^9$
T (7 per Toffoli)	$1.28 \cdot 10^{10}$
qubits	204,049
Part II search, $L = 2n = 512$	
target fraction q	$7/\sqrt{p} = 2^{-124.9}$
Grover iterations	$2^{62.1}$
Toffoli	$2^{102.9}$
two-qubit, $\{cx, rz, sx, x\}$	$2^{105.9}$
depth	$2^{104.4}$
logical qubits	466,000
classical random routes	$2^{133.9}$ steps

Part III: trade-offs, other algorithms



Each row changes one algorithm of the submitted circuit:

change	Toffoli	factor
submitted	139,236	1.00
root by powers in $\mathbb{F}_{p^2}$	339,162	2.44
+ $\mathbb{F}_{p^2}$ product as 4 products	380,506	2.73
+ flag by subtract and add back	431,914	3.10

Root by powers in $\mathbb{F}_{p^2}$: Algorithm 9 of Adj and Rodríguez-Henríquez (2014).

- The root is 86% of the step. In $\mathbb{F}_p$ a product is 1 MAC; in $\mathbb{F}_{p^2}$ it is 3 or 4.
- A comparison needs only the carry: $2n$ Toffolis, not $4(n+1)$.
- At $n = 256$: submitted $= 0.34\times$ the slowest version.

G. Adj, F. Rodríguez-Henríquez, *Square root computation over even extension fields*, IEEE Trans. Computers 63 (2014).

Part III: trade-offs, qubits and depth

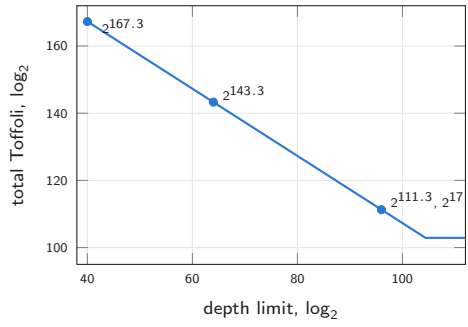
Qubits: which values to keep (pebbling)

Bennett's method keeps each intermediate value: about $1.5n$ registers of n qubits per exponentiation, so qubits $\sim n^2$.

strategy	registers	steps run
Bennett (ours)	T	$2T$
$\sqrt{T}$ checkpoints	$2\sqrt{T}$	$\approx 3T$
recursive halving	$\log_2 T$	$T^{1.58}$

We keep Bennett: it gives the fewest gates.

Part II search at $p \approx 2^{256}$ with a depth limit



With depth limit 2^m: $S = (2^{104.4-m})^2$ machines; total $\times\sqrt{S}$.

C. H. Bennett, *Logical reversibility of computation*, IBM J. Res. Dev. 17 (1973).
C. H. Bennett, *Time/space trade-offs for reversible computation*, SIAM J. Comput. 18 (1989).

Part IV: hidden names (Djinns)

The spot names are replaced by random labels. The Djinn returns the label at the end of a road; the volunteer reports whether a labelled spot is on the Clock Tower or at ANC.

What still works

- The graph is the same: degree, roads, cycles, expansion.
- Classical: random walks ($p^{1/2}$ tower tests), breadth- or depth-first search with a table of labels, cycle and collision search.
- Our Part I method: the survey asks the Djinn for every road ($O(N)$), then the step circuit is built from the survey.
- If the Djinn and volunteer accept superpositions (coherent access), Grover and quantum walks still give $p^{1/4}$.

What stops working

- Arithmetic on A: the Part III step circuit, the j test, the Frobenius map $a + bi \mapsto a - bi$.
- Methods from number theory (class groups, endomorphism rings).
- A Djinn or volunteer that answers classically takes one question at a time. No superposition, so no Grover.

Lower bound

The names are random, so the volunteer is close to a black box. The $\Omega(p^{1/4})$ bound needs one more condition: the tower set must not be visible from the graph shape.

F. Magniez, A. Nayak, J. Roland, M. Santha, *Search via quantum walk*, SIAM J. Comput. 40 (2011).

Summary

- **Part I:** all 37 passwords, k counted at confidence ≥ 0.99, survey rules kept.
- **Part II:** $p^{1/4+o(1)}$ quantum against $p^{1/2+o(1)}$ classical. Black-box lower bound; a lower bound for the real arithmetic would prove P $\neq$ NP. Fitted exponent 0.523 ± 0.040; Grover demo $=$ theory.
- **Part III:** one step from arithmetic only, 510/510 correct. At 2^{256}: $1.8 \cdot 10^9$ Toffolis per step, 2^{103} for the search.
- **Part IV:** the speed-up needs oracles that accept superpositions, not the spot names.

Reproduce

```
uv run oasis --part 1 --p 1019 \
  --A0 "977 + 297i" --L 12 \
  --jstar "797 + 921i"
uv run oasis --part 2
uv run oasis --part 3
```

Results: `results/`

A1. Iterative Amplitude Estimation

- Problem: $A|0\rangle = \sqrt{a}\,|\text{good}\rangle + \sqrt{1-a}\,|\text{bad}\rangle$. Estimate $a = \sin^2\theta_a$.
- Grover operator $Q = AS_0A^\dagger S_{\text{good}}$, where S_0 and S_{good} flip the sign of $|0\rangle$ and of the good states. Then $Q^mA|0\rangle$ is good with probability $\sin^2((2m+1)\theta_a)$.
- Each round picks the largest power m that keeps $(2m+1)\theta_a$ inside one half-period of the current interval. It measures, and narrows the interval for θ_a (Clopper–Pearson: an exact confidence interval for a binomial proportion).
- Cost: $O\big(\frac{1}{\varepsilon}\log\frac{2}{\alpha}\log\frac{\pi}{4\varepsilon}\big)$ queries for error ε at confidence $1-\alpha$, with no QFT (Grinko, Gacon, Zoufal, Woerner 2021).
- Here: good = spot register at an ANC code, $a = k/N$. We halve ε until the interval for $k = aN$ holds one integer; $\alpha = 0.01$.

D. Grinko, J. Gacon, C. Zoufal, S. Woerner, *Iterative quantum amplitude estimation*, npj Quantum Inf. 7 (2021).
C. J. Clopper, E. S. Pearson, *The use of confidence or fiducial limits illustrated in the case of the binomial*, Biometrika 26 (1934).

A2. Theorems we use

Deuring–Eichler For $p \equiv 3 \pmod 4$, the number of supersingular $j \in \mathbb{F}_p$ is $2h(-p)$ if $p \equiv 3 \pmod 8$ and $h(-p)$ if $p \equiv 7 \pmod 8$.

Siegel $h(-p) = p^{1/2+o(1)}$.

Pizer The supersingular 2-isogeny graph is 3-regular and Ramanujan: every nontrivial eigenvalue has $|\lambda| \leq 2\sqrt{2}$.

Lubetzky–Peres For a walk without backtracking of length L on a 3-regular Ramanujan graph with N_E directed edges: $\left\|\tilde{\mu}_L/\pi_E - 1\right\|_{L^2}^2 \leq 2N_E 2^{-L}(8L^2+1)$.

BBBV, Zalka Search for one of k marked items among N with a black-box oracle needs $\Omega(\sqrt{N/k})$ queries; Grover is optimal.

BBHT With k unknown and $\lambda = 6/5$, the expected number of Grover iterations is at most $\frac{9}{4}\sqrt{N/k}$.

M. Deuring, *Die Typen der Multiplikatorenringe elliptischer Funktionenkörper*, Abh. Math. Sem. Hamburg 14 (1941).
C. Delfs, S. D. Galbraith, *Computing isogenies between supersingular elliptic curves over $\mathbb{F}_p$*, Des. Codes Cryptogr. 78 (2016).
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A3. The Clock-Tower problem is in NP

Decision problem: given p and A_0, is there a route of length $L = O(\log p)$ that ends on the Clock Tower?

1. **Certificate:** the route $x = b_1 \cdots b_L$. Its length is $O(\log p)$ bits.
2. **Check:** L steps, each a fixed number of operations in $\mathbb{F}_{p^2}$ (one square root is one exponentiation), then compute j and test $\mathrm{Im}\, j = 0$. Each operation costs $\mathrm{poly}(\log p)$, so the check costs $\mathrm{poly}(\log p)$.
3. So the problem is in NP. The input size is $n = \log_2 p$.
4. Our cost $p^{1/4} = 2^{n/4}$ is exponential in n. A proof that no polynomial-time quantum algorithm exists would show NP $\not\subseteq$ BQP. Since P $\subseteq$ BQP, this gives P $\neq$ NP.
5. So an unconditional lower bound for the real arithmetic is out of reach today. Lower bounds exist only in a model with a black-box oracle.

A4. Known algorithms that use the arithmetic

algorithm	solves	cost	why not here
Biasse–Jao–Sankar (2014)	path between two curves, quantum	$\widetilde{O}(p^{1/4})$	same exponent as Grover
Page–Wesolowski (2023)	path, classical, from both ends	$\widetilde{O}(p^{1/2})$	meet-in-the-middle; forbidden by the rules
Delgado (2026)	one endomorphism, classical	$p^{1/3+o(1)}$	not a fixed-degree path; the path version needs GRH and gives longer paths
Wesolowski (2021)	path $\Leftrightarrow$ endomorphism ring	reduction	assumes GRH

None of these gives less than $p^{1/4}$ for a route of length $\Theta(\log p)$ from A_0 under the rules.

GRH: the generalized Riemann hypothesis. Endomorphism ring: the ring of isogenies from a curve to itself. Meet-in-the-middle: search from both ends and match.

J.-F. Biasse, D. Jao, A. Sankar, *A quantum algorithm for computing isogenies between supersingular elliptic curves*, INDOCRYPT 2014.
A. Page, B. Wesolowski, *The supersingular Endomorphism Ring and One Endomorphism problems are equivalent*, arXiv:2309.10432 (2023).
J. L. Delgado, *The Supersingular Isogeny Problem in Time and Memory $p^{1/3+o(1)}$, Unconditionally*, arXiv:2609.22018 (2026).
B. Wesolowski, *The supersingular isogeny path and endomorphism ring problems are equivalent*, arXiv:2111.01481 (2021).

A5. Proof of the square root from the norm

Let $s = x_0 + x_1 i$ with $s^2 = D = \alpha + \beta i$. Then $x_0^2 - x_1^2 = \alpha$, $2x_0x_1 = \beta$, and $(x_0^2 + x_1^2)^2 = \alpha^2 + \beta^2 = m$.

- m is a square in $\mathbb{F}_p$, so $\lambda = m^{(p+1)/4}$ gives $\lambda^2 = m$, and $x_0^2 + x_1^2 = \pm\lambda$.
- $\delta = (\alpha + \lambda)/2$ and $\delta' = (\alpha - \lambda)/2$ satisfy $\delta\delta' = -\beta^2/4$. Since -1 is not a square ($p \equiv 3 \bmod 4$), exactly one of them is a square (if $\beta \neq 0$).
- $r = \delta^{(p-3)/4}$, $y = r\delta$: then $y^2 = \delta^{(p+1)/2} = \pm\delta$ and $yr = \delta^{(p-1)/2} = \pm 1 = q$-flag.
- $q = 1$: $y^2 = \delta = x_0^2$, $x_1 = \beta/(2y) = \beta r/2 = t$, so $s = y + ti$.
- $q = 0$: $y^2 = -\delta = x_1^2$, $x_0 = \beta/(2y) = -\beta r/2 = -t$, so $s = -t + yi$.
- $\delta = 0$ only if $\beta = 0$ and α is not a square; then $\delta := \alpha$ gives $q = 0$, $s = yi$ with $y^2 = -\alpha$.

Checked on every nonzero square for 13 primes ≤ 83, and in the circuit on random squares at $p = 43$ and $p = 131$.

A6. Resources

Part I, $p = 1019$, $L = 12$, $\{cx, rz, sx, x\}$, level 1

	queries	CX	depth
one step	1	13,395	42,173
Grover iteration	24	319,106	1,009,282

22 qubits. Total: 854,784 queries to count, 26,304 to search.

Part II, $p = 5923$, $L = 12$, $\{cx, rz, sx, x\}$, level 1

	queries	CX	depth
one step	1	141,813	462,568
Grover iteration	24	3,401,466	11,100,684
search, $t = 4$	96	13,605,864	44,402,739

25 qubits.

Part III, $p = 1019$, $\{x, cx, ccx\}$; two-qubit in $\{cx, rz, sx, x\}$ = CX +6·Toffoli

	qubits	Toffoli	2-qubit	depth
step	587	139,236	1,127,164	307,933
tower test	192	37,794	–	63,148

T: 974,652 (7 per Toffoli) or 556,944 (4 per Toffoli).

Part III, $n = 256$

step: Toffoli / qubits	$1.83 \cdot 10^9$ / 204,049
tower test: Toffoli	$2.1 \cdot 10^7$
search: Toffoli / depth	$2^{102.9}$ / $2^{104.4}$

A7. Note on the phase test

- The test prepares $\cos w\,|0\rangle\,|x\rangle + e^{i\phi}\sin w\,|1\rangle\,|y\rangle$, runs the new circuit, then the inverse of the reference circuit, then undoes the preparation. The result is $|0\rangle$ with probability $1 - \sin^2(2w)\sin^2(\Delta\theta/2)$.
- A path of pairs through the inputs gives $\theta(x) = \theta(y)$ for all of them: the two circuits are equal up to one global phase.
- With equal amplitudes ($w = \pi/4$), Aer's matrix-product-state simulator lost the relative phase for some values of ϕ, also for a circuit followed by its own inverse. Equal amplitudes give two equal Schmidt coefficients (the singular values across a cut of the state), so the simulator's singular value decomposition has no unique answer.
- With $w = 0.55$ every case passes. A deliberate error (Z on b) is found on every pair with different b, and only there.

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